Diego Alejandro Clavijo Ladino

Junior Frontend React Developer

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PROFESSIONAL SUMMARY

Systems and Computer Engineer specializing in frontend development with React and TypeScript. I design maintainable, scalable, quality-driven software solutions, applying sound engineering practices to build reliable, user-centered products. I communicate technical decisions clearly and collaborate actively within development teams, both in independent projects and institutional team-based work.

PROFESSIONAL EXPERIENCE

Independent Software Engineering Projects *September 2022 - Present*

Junior Frontend React Developer

- Designed modular React architecture with layered separation of concerns and an adapter pattern in the data layer, applied across 2 full-stack applications (Cocktail Lab in production, Streamline actively in development).
- Implemented state management with Zustand and runtime schema validation with Zod, ensuring strict end-to-end type safety between frontend and API.
- Designed a 5-stage, multi-layer testing strategy (1,003 tests across 42 files, 100% passing in CI) with 89% line coverage — unit tests (Vitest + React Testing Library), accessibility audits (jest-axe), integration testing with network mocking (MSW), and 500 cross-browser E2E tests across 5 engines (Chromium, Firefox, WebKit, Mobile Chrome, Mobile Safari) with Playwright.
- Integrated an AI-powered recipe generator (Groq LLM) as a Vercel Serverless Function, with runtime schema validation (Zod) and image resolution via a third-party API integration, including fallback handling for failures.
- Improved accessibility through ARIA attributes and keyboard navigation, meeting WCAG 2.1 AA compliance validated with axe-core.
- Automated CI pipelines with GitHub Actions, documenting architecture and technical decisions in engineering READMEs.

Fundación de Levi *February 2022 - August 2022*

Web Developer

- Designed the gallery section of Fundación de Levi's institutional website, defining its visual layout and navigation structure as part of a team-based development project.
- Developed the gallery section using HTML, CSS, and JavaScript, implementing a responsive, cross-browser-compatible design.

PROJECTS

Cocktail Lab - Production | React, TypeScript, Vitest, Playwright, MSW, jest-axe, Groq AI

React application with scalable frontend architecture, generative AI integration, and a multi-layer testing strategy (1,003 tests, 89% coverage) with an emphasis on WCAG 2.1 AA accessibility.

https://github.com/DAlejandroCl/Cocktail-Lab | https://cocktail-lab-devacl.vercel.app

Streamline Inventory - Actively in development | PERN Stack, Docker, GitHub Actions

Full-stack inventory management application with clean architecture, layered separation of concerns, and strict typing in a pnpm + Turborepo monorepo, built to demonstrate enterprise-level engineering practices. CI/CD pipeline and E2E test suite currently under active development.

https://github.com/DAlejandroCl/streamline-inventory

PUBLICATIONS

Security Model with a Holistic Approach for Mobile Applications of Financial Entities in Colombia

IEEE, 2025

Co-authored a holistic security framework — grounded in recognized standards and frameworks — for assessing and mitigating the risks of financial mobile app adoption in the Colombian context, addressing both technical and human factors beyond the traditional user-centric narrative.

EDUCATION

B.Sc.: Systems and Computer Engineering *2018 - 2024*

Universidad Católica de Colombia

TECHNICAL SKILLS

TypeScript, JavaScript (ES6+), HTML5, CSS3, React, Zustand, Zod, Vite, WCAG 2.1 AA, Node.js, REST APIs, PostgreSQL, Vitest, React Testing Library, Playwright, MSW, jest-axe, Git, GitHub Actions, Docker, SCRUM, Spanish (Native), English (B2)

CERTIFICATIONS

React & Full-Stack Development (MERN/PERN) - Udemy, In progress, 2026 | AI from 0 to Production - BIG School, 2026 | Advanced CSS and SASS - Udemy, 2025